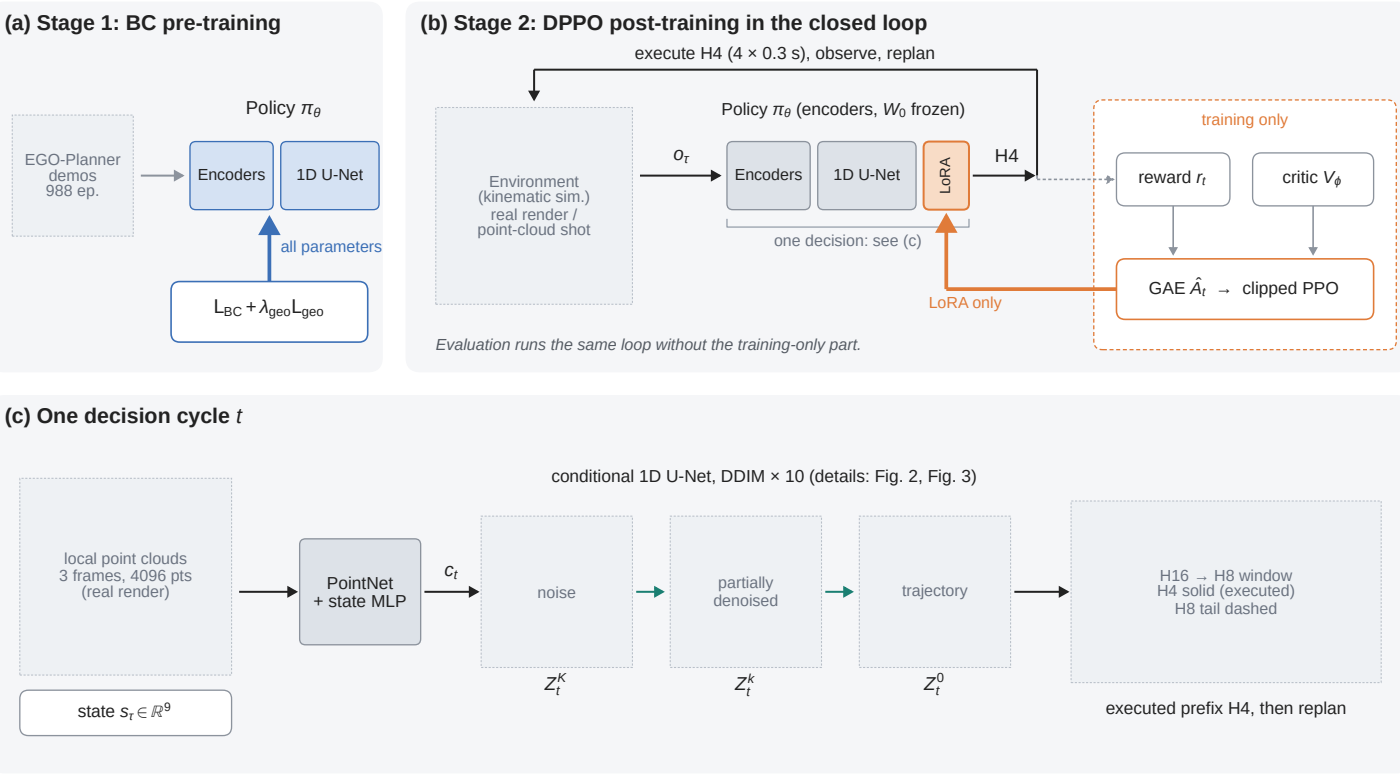
(a) Stage 1: BC pre-training
Policy π_θ
EGO-Planner demos 988 ep.
Encoders
1D U-Net
all parameters
$L_{BC} + \lambda_{geo} L_{geo}$
(b) Stage 2: DPPO post-training in the closed loop
execute H4 (4 × 0.3 s), observe, replan
Environment (kinematic sim.) real render / point-cloud shot
Policy π_θ (encoders, W_0 frozen)
o_τ
Encoders
1D U-Net
LoRA
H4
one decision: see (c)
training only
reward r_t
critic V_ϕ
GAE $\hat{A}_t$ → clipped PPO
LoRA only
Evaluation runs the same loop without the training-only part.
(c) One decision cycle t
conditional 1D U-Net, DDIM × 10 (details: Fig. 2, Fig. 3)
local point clouds 3 frames, 4096 pts (real render)
state $s_\tau \in \mathbb{R}^9$
PointNet + state MLP
c_t
noise
Z_t^K
partially denoised
Z_t^k
trajectory
Z_t^0
H16 → H8 window
H4 solid (executed)
H8 tail dashed
executed prefix H4, then replan